```
void gemm  (int N, float *A, float *B, float C) {
    int i, j;
    float sum=0.0;
    #pragma HLS pipeline II=1
    for(int i = 0; i < N; i++){
        sum = 0.0;
        #pragma HLS unroll factor=4
        for(j = 0; j < 4; j++){
                sum += C * A[i*4+j];
            }
        B[i] = sum;}
}
```